42057 Introduction to Space Communications and Sensing

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Binary Data and Spread-Spectrum Communications

Assignment 2 — Topic 2

UTS Faculty of Engineering & IT

Student: [Student Name] ([Student ID])
Group: Project 2 Group 1
Members: Jaylen Avtarovski, Benjamin Hudson, Rhyse Williams, Elina Kim
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Abstract

A baseband binary communication system and its direct-sequence spread-spectrum (DSSS) extension were designed, implemented in MATLAB and evaluated by Monte-Carlo simulation. Each block of the textbook system models (bit source, rectangular-pulse modulator, uniform noise channel, integrate-and-dump demodulator and threshold detector) is a separate MATLAB function, and the same functions are reused for the spread-spectrum system by adding a ± 1 pseudo-noise (PN) sequence and a sinusoidal jammer. For the unjammed system the measured bit error rate (BER) agrees with the exact closed-form (Irwin–Hall) result at every simulated point: the exact value lies inside the 95% confidence interval of all 26 estimates, which span BERs from about 0.2 down to 5×10^{-8} (the lowest points rest on few errors and carry wide intervals). The common Gaussian approximation $\mathcal{Q}(\sqrt{\mathrm{SNR}})$ is pessimistic at high SNR because uniform noise is bounded. For the jammed system, despreading converts the narrowband jammer into wideband noise. The ideal processing gain is $10 \log_{10} M$ (17 dB to 30 dB); the measured improvement in the jamming power tolerated at BER $= 10^{-3}$ was 11.4 dB to 20.5 dB for $M = 50$ to 1000. Under the brief's literal condition $A = M$ the outcome depends on the jammer frequency. Against an in-band jammer both systems operate near BER $= 0.5$, because the jammer power grows as M^2 while the processing gain grows only as M, and the PN system is only modestly better. Against an out-of-band jammer the conventional receiver's own low-pass response rejects the tone (BER 0 for $M = 50$ and 100, 0.013 for $M = 1000$, but 0.33 for $M = 500$), while the PN system stays near 0.42–0.48 and is worse. The jammed results agree with semi-analytical expressions (Gaussian treatment of the summed noise, averaged over the jammer phase) derived in the report.

Contents

1 Introduction **2**

2 Theory **2**
2.1 Binary system with additive noise . . . 2
2.2 Sinusoidal jamming without spreading . . . 3
2.3 Direct-sequence spreading . . . 4

3 MATLAB implementation **4**
3.1 Structure . . . 4
3.2 Simulation design . . . 5

4 Results: Project 12.7 **5**

5 Results: Project 12.8 **6**
5.1 The brief's experiment: $A = M$ . . . 6
5.2 Effect of jammer power . . . 8
5.3 Effect of jammer frequency . . . 9
5.4 Spectral view . . . 9

6 Discussion **10**
6.1 Effect of the PN sequence on the sinusoidal interference . . . 10
6.2 Why the spread-spectrum system outperforms the conventional one . . . 10
6.3 Interpreting the condition $A \geq M$ . . . 11
6.4 Limitations . . . 11

7 Conclusion **12**

Individual contribution **12**

A MATLAB code **12**

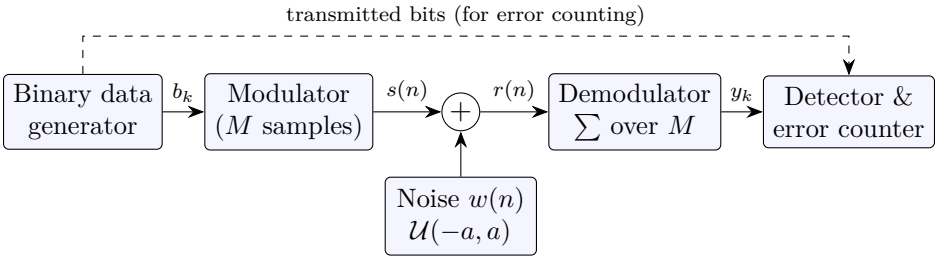


Figure 1: Binary data communication system of Project 12.7 (after Fig. 12.16 of [1]).

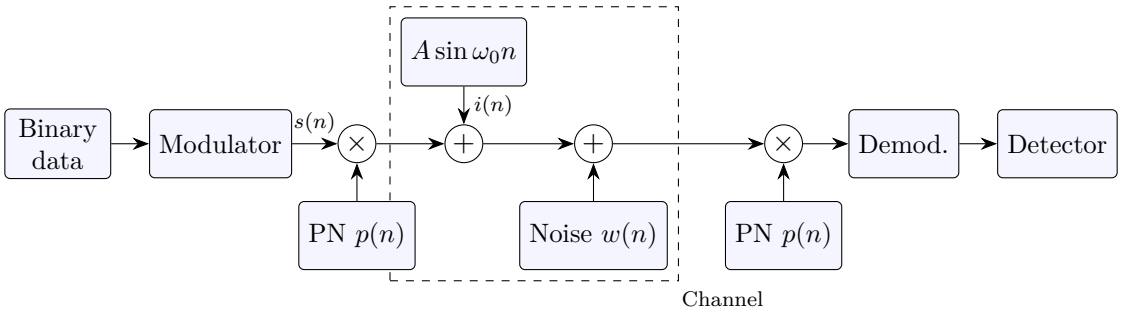


Figure 2: Binary PN spread-spectrum system of Project 12.8 (after Fig. 12.18 of [1]). The same PN sequence is used at the transmitter and the (synchronised) receiver.

1 Introduction

Spacecraft links are power limited, operate over long distances, and are exposed to both unintentional radio-frequency interference and deliberate jamming. Spread spectrum is the standard answer to the second problem: the Global Positioning System spreads its 50 bit/s navigation message with a 1.023 Mchip/s C/A code [4], a processing gain of about 43 dB, and NASA's Tracking and Data Relay Satellite System uses direct-sequence spreading on its multiple-access links [5] for the same reasons: interference rejection, low power spectral density and code-division multiple access.

This assignment studies the mechanism behind that protection using two projects from Ingle and Proakis [1]:

- **Project 12.7** — a binary data communication system over an additive uniform-noise channel (Figure 1); the BER is measured as a function of the signal-to-noise ratio $\mathrm{SNR} = M/P_n$.
- **Project 12.8** — the same system with the modulator output multiplied by a binary PN sequence and a channel that also adds a sinusoidal jammer $i(n) = A \sin \omega_0 n$ with $A \geq M$ (Figure 2); the BER is measured with and without the PN sequence for $M = 50, 100, 500, 1000$.

The objectives were to (i) implement every block of both systems as a reusable MATLAB function, (ii) verify the simulator against theory, (iii) quantify the effect of the PN sequence on the jammer, and (iv) explain why, and when, the spread-spectrum system outperforms the conventional one. Section 2 develops the theory, Section 3 describes the implementation, Sections 4 and 5 present the results, and Section 6 discusses them.

2 Theory

2.1 Binary system with additive noise

Bit $b_k \in \{0, 1\}$ is mapped to the antipodal level $d_k = 2b_k - 1 = \pm 1$ and repeated for M samples,

$$s(n) = d_k, \qquad kM \leq n < (k+1)M. \tag{1}$$

The channel adds i.i.d. noise $w(n) \sim \mathcal{U}(-a, a)$ with power $P_n = \mathrm{E}[w^2] = a^2/3$. The demodulator sums the M received samples of each bit,

$$y_k = \sum_{n=kM}^{(k+1)M-1} r(n) = M d_k + \underbrace{\sum_{n} w(n)}_{W_k}, \tag{2}$$

which is the matched filter (integrate-and-dump) for a rectangular pulse. The detector decides $\hat{b}_k = 1$ if $y_k > 0$. The signal component has energy M^2 and the noise component W_k has variance MP_n, so the post-detection SNR is $M^2/(MP_n) = M/P_n$, which is the definition used in the brief:

$$\mathrm{SNR} = \frac{M}{P_n}. \tag{3}$$

By symmetry the bit error probability is $P_e = \Pr\{W_k > M\}$.

Gaussian approximation. For large M the central limit theorem gives $W_k \approx \mathcal{N}(0, MP_n)$ and

$$P_e \approx \mathcal{Q}\Big(\frac{M}{\sqrt{MP_n}}\Big) = \mathcal{Q}\Big(\sqrt{\mathrm{SNR}}\Big), \qquad \mathcal{Q}(x) = \tfrac{1}{2}\operatorname{erfc}(x/\sqrt{2}). \tag{4}$$

Exact result. W_k is a sum of M independent uniform variables, so $W_k = 2aS - Ma$ where S is a sum of M independent $\mathcal{U}(0,1)$ variables, which has the Irwin–Hall distribution. Hence

$$P_e = \Pr\{S > x\} = F_M(M - x), \qquad x = \frac{M(1+a)}{2a}, \qquad F_M(y) = \frac{1}{M!}\sum_{k=0}^{\lfloor y \rfloor} (-1)^k \binom{M}{k} (y-k)^M, \tag{5}$$

using the symmetry of S about $M/2$. Evaluating F_M on the short side ($y \le M/2$) keeps the alternating sum well conditioned; for the $M \le 20$ used here the double-precision values agree with a 50-digit evaluation to at least seven significant figures (`exact_ber_uniform.m`). Two special cases are useful checks: for $M = 1$, $P_e = (a-1)/(2a)$ for $a > 1$; and for any M, $P_e = 0$ when $a \le 1$ because $|W_k| \le Ma \le M$. Because the true density has bounded support, its tails fall off faster than the Gaussian, so (4) overestimates P_e at high SNR.

2.2 Sinusoidal jamming without spreading

With the jammer present, $y_k = Md_k + W_k + J_k$, where

$$J_k = A \sum_{n=kM}^{kM+M-1} \sin \omega_0 n = A \underbrace{\frac{\sin(M\omega_0/2)}{\sin(\omega_0/2)}}_{D_M(\omega_0)} \sin\Big(\omega_0\big(kM + \tfrac{M-1}{2}\big)\Big). \tag{6}$$

D_M is the Dirichlet kernel: the integrate-and-dump demodulator is a low-pass filter with gain M at DC, main lobe $|\omega_0| < 2\pi/M$ and nulls at $\omega_0 = 2\pi\ell/M$. Two regimes follow.

- **In-band jammer** ($\omega_0 \ll 2\pi/M$): $D_M \approx M$, so the jammer is integrated exactly like the signal, $|J_k|$ reaches AM, and an error occurs whenever $|A\sin(\cdot)| > 1$. Increasing M does *not* help.
- **Out-of-band jammer**: $|J_k| \le A/|\sin(\omega_0/2)|$, independent of M, so the demodulator already rejects it unless A is very large.

Averaging over the jammer phase ϕ at the start of a bit (uniform for incommensurate ω_0) and treating W_k as Gaussian,

$$P_e^{\text{no PN}} = \mathrm{E}_\phi\Big[\tfrac{1}{2}\mathcal{Q}\Big(\frac{M + J(\phi)}{\sqrt{MP_n}}\Big) + \tfrac{1}{2}\mathcal{Q}\Big(\frac{M - J(\phi)}{\sqrt{MP_n}}\Big)\Big]. \tag{7}$$

2.3 Direct-sequence spreading

The transmitter sends $s(n)p(n)$, where $p(n) = \pm 1$ are equiprobable i.i.d. chips. The receiver multiplies by the same $p(n)$; since $p^2(n) = 1$,

$$r(n)p(n) = s(n) + w(n)p(n) + A\sin(\omega_0 n)\,p(n). \tag{8}$$

The desired signal is restored exactly, the noise is statistically unchanged, and the jammer is multiplied by a random ± 1 sequence. Its spectrum is therefore spread uniformly over $(-\pi, \pi]$: the jammer becomes zero-mean white noise of power $A^2/2$. After summing over a bit,

$$J'_k = A\sum_n p(n)\sin\omega_0 n, \qquad \mathrm{E}[J'_k] = 0, \qquad \mathrm{Var}[J'_k] = A^2\sum_n \sin^2\omega_0 n \approx \frac{MA^2}{2}, \tag{9}$$

and the signal-to-interference-plus-noise ratio at the detector is

$$\mathrm{SINR} = \frac{M^2}{MP_n + MA^2/2} = \frac{M}{P_n + A^2/2}, \qquad P_e^{\mathrm{PN}} \approx \mathcal{Q}\left(\sqrt{\frac{M}{P_n + A^2/2}}\right). \tag{10}$$

Comparing with the per-sample signal-to-jammer ratio $2/A^2$ shows a *processing gain* $G_p = M$ ($10\log_{10} M$ dB): spreading lets the integrator average the jammer down in exactly the way it averages white noise. For very low ω_0 the factor $\sum\sin^2$ differs from bit to bit, so the simulations are compared with the phase-averaged form $\mathrm{E}_\phi[\mathcal{Q}(M/\sqrt{MP_n + A^2\sum\sin^2(\omega_0 n + \phi)})]$ (`ber_theory_jam.m`).

Consequence for the condition $A \geq M$. The jammer-to-signal power ratio is $\mathrm{JSR} = A^2/2$, so with $A = M$ it is $M^2/2$ while the processing gain is only M. Equation (10) then gives $\mathrm{SINR} \approx 2/M$, which *decreases* with M: under the brief's condition the PN system cannot reach a low error rate at any jammer frequency. The conventional system cannot either when the jammer is in band, but an out-of-band jammer may still be rejected by its integrator, depending on $|D_M(\omega_0)|$ in (6). These predictions are confirmed in Section 5.

3 MATLAB implementation

3.1 Structure

Each block of Figures 1 and 2 is a separate function in the `matlab/` folder (Table 1); the five functions required by Project 12.7 are reused unchanged in Project 12.8. The code uses only core MATLAB (no toolboxes) and also runs in GNU Octave, which was used to produce the results in this report. Full listings are in Appendix A.

The core of the simulator, `binary_comm_sim`, follows the block diagram line by line:

```
b = bit_generator(Nb); % binary data generator
s = modulator(b, M); % modulator
if use_pn
    p = pn_generator(L); % PN sequence generator
    s = s .* p; % spreading
end
r = s + noise_generator(L, a); % channel: wideband noise
if A ~= 0
    r = r + sine_generator(L, A, w0, n0); % channel: sine jammer
end
if use_pn
    r = r .* p; % despreading
end
y = demodulator(r, M); % demodulator
[~, e] = detector(y, b); % detector / error counter
```

Table 1: MATLAB functions and the block each implements.

Function	Block	Operation
`bit_generator(N)`	Binary data generator	$b_k = [u_k \geq 0.5]$, $u_k \sim \mathcal{U}(0,1)$
`modulator(b,M)`	Modulator	repeat $2b_k - 1$ for M samples
`noise_generator(L,a)`	Noise generator	$a(2u-1)$, uniform on $(-a, a)$
`demodulator(r,M)`	Demodulator	column sums of an $M \times N$ reshape
`detector(y,b)`	Detector & counter	$\hat{b}_k = [y_k > 0]$, count $\hat{b}_k \neq b_k$
`pn_generator(L)`	PN generator	equiprobable ± 1 chips
`sine_generator(L,A,w0,n0)`	Sine generator	$A \sin \omega_0 n$, continuous phase
`binary_comm_sim`	Whole system	chains the blocks, Monte-Carlo loop
`exact_ber_uniform`	Theory	exact P_e, Irwin–Hall (5)
`ber_theory_jam`	Theory	Equations (7) and (10), phase averaged

3.2 Simulation design

- **Vectorisation and memory.** Bits are processed in blocks of up to 2×10^7 samples so that $M = 1000$ runs do not exhaust memory; the jammer sample index n_0 is carried between blocks so its phase is continuous.

- **Stopping rule.** For Project 12.7 each SNR point runs until 200 errors are counted or 2×10^7 bits are simulated. With 200 errors the relative standard error of the BER estimate is about $1/\sqrt{200} \approx 7\%$. Points that hit the bit limit first have fewer errors and are less certain, so every Project 12.7 estimate is reported with its error count and a 95% Clopper–Pearson (exact binomial) interval (`ber_ci.m`). For the $A = M$ experiment a fixed budget of 2×10^7 samples per point (4×10^5 bits at $M = 50$, 2×10^4 bits at $M = 1000$) was used; at BER ≈ 0.45 this gives a standard error below 0.004.

- **Parameters.** Unless stated otherwise $P_n = 1$ ($a = \sqrt{3}$). Two jammer frequencies were used: $\omega_0 = 0.002$ rad/sample, inside the demodulator pass-band $2\pi/M$ for every M tested, and $\omega_0 = 1$ rad/sample ($\approx 0.318\pi$), outside it. Both are deliberately incommensurate with $2\pi/M$; a frequency such as 0.1π with $M = 100$ places the jammer exactly on a null of D_M and gives a misleading zero error rate.

- **Reproducibility.** Seeds are fixed with `rng`; `run_all.m` regenerates every number and figure, and the tables in this report are read directly from the files the scripts write.

4 Results: Project 12.7

Figure 3 traces eight bits through the system at $M = 10$, $P_n = 1$ (SNR = 10 dB). The received samples are individually unreliable (many have the wrong sign), but summing M of them gives decision statistics close to ± 10 and all eight bits are detected correctly.

The BER was measured in the two ways suggested in the brief: (a) $P_n = 1$ fixed with $M = 1, \ldots, 20$, and (b) $M = 10$ fixed with P_n varied to give SNR = 0–13 dB. Figure 4 and Table 2 show the results.

Observations:

1. The BER falls steeply with SNR, from about 0.2 at 0 dB to 10^{-6} near 13 dB; each extra dB costs roughly a factor of three to ten in BER at high SNR.

2. The exact value lies inside the 95% interval at all 26 simulated points of both sweeps, which supports the implementation of all five blocks. At $M = 1$ the simulated 0.21 matches $(a-1)/(2a) = 0.211$. The agreement is tightest where errors are plentiful; above about 12 dB

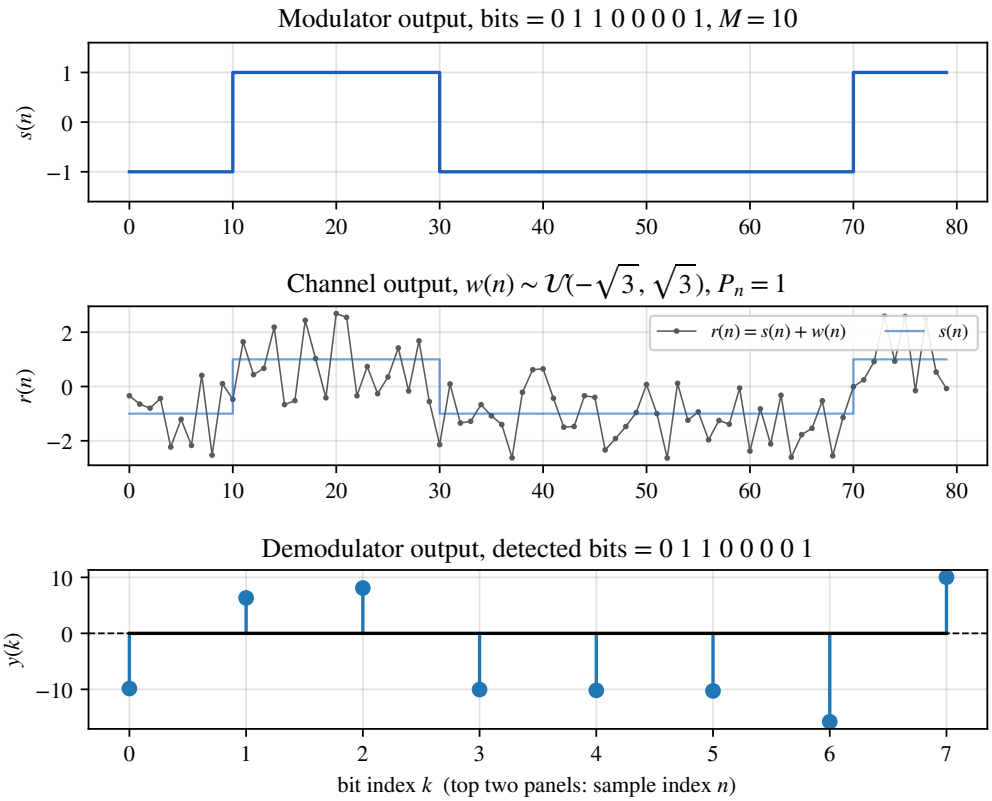


Figure 3: Signals at the modulator output, channel output and demodulator output for $M = 10$ and $P_n = 1$.

the bit budget limits the error count (83 and 28 errors at $M = 18$ and 20), so their 95% intervals are about ±20% and ±40% wide.

3. The Gaussian approximation is accurate between about 3 and 8 dB but is optimistic at $M = 1$ (a single uniform sample has heavier mass near its edges than a Gaussian) and increasingly pessimistic at high SNR, by a factor of three at 13 dB in sweep (a).

4. The two sweeps coincide at low SNR, as the definition SNR $= M/P_n$ implies, but separate at high SNR: for sweep (b) with $M = 10$ fixed, high SNR means a small noise range a, and as $a \to 1$ the maximum possible noise sum Ma approaches M, so $P_e \to 0$ much faster than the Gaussian tail. At 13 dB only one error occurred in 2×10^7 bits. The point estimate 5×10^{-8} is therefore very uncertain: its 95% interval is $[1.3 \times 10^{-9}, 2.8 \times 10^{-7}]$, which contains the exact value 1.2×10^{-7} but would also be consistent with values more than an order of magnitude lower, so this point should not be read as a precise measurement. The SNR alone does not fully describe the system when the noise is not Gaussian.

5 Results: Project 12.8

5.1 The brief's experiment: $A = M$

Table 3 and Figure 5 give the BER with and without the PN sequence for $M = 50, 100, 500, 1000$ with the smallest jammer amplitude the brief allows, $A = M$.

- **In-band jammer.** Without PN the BER is 0.49–0.50 for every M: the jammer passes through the integrator with the full gain M, so the bit sum is dominated by $AM \sin(\cdot)$ and the decision is essentially a coin toss. With PN the BER is lower, 0.37 at $M = 50$, because

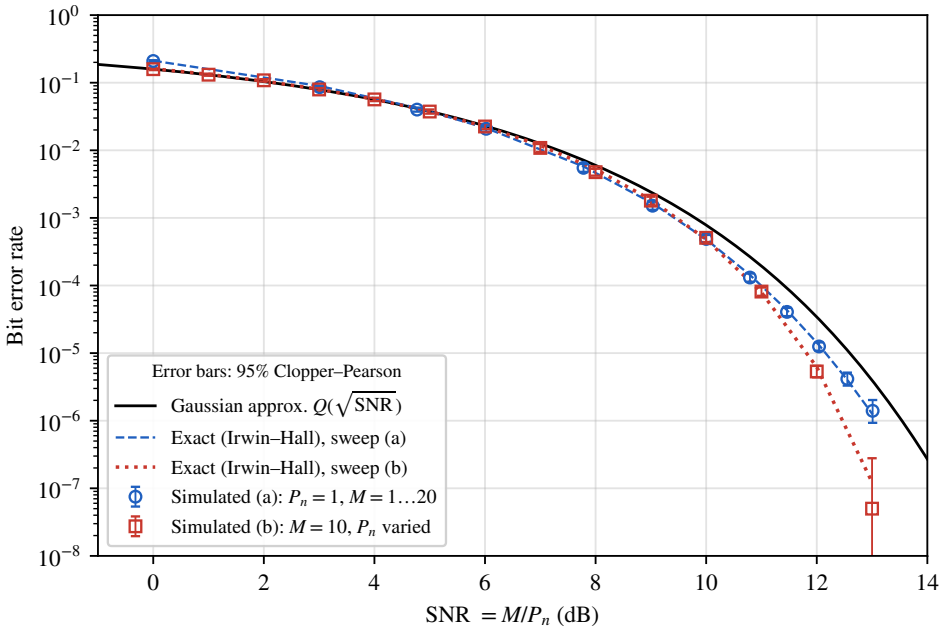


Figure 4: BER against SNR $= M/P_n$ for the binary system with uniform noise.

Table 2: Sweep (a): $P_n = 1$, M varied. Simulated BER with its error count and 95% Clopper–Pearson interval, compared with the exact probability (5) and the Gaussian approximation $\mathcal{Q}(\sqrt{\text{SNR}})$.

M	SNR (dB)	Errors	Bits	Simulated	95% interval	Exact	$\mathcal{Q}(\sqrt{\text{SNR}})$
1	0.00	4190	2.0×10^4	0.2095	[0.2039, 0.2152]	0.2113	0.1587
2	3.01	1717	2.0×10^4	0.0858	[0.0820, 0.0898]	0.0893	0.0786
3	4.77	800	2.0×10^4	0.0400	[0.0373, 0.0428]	0.0425	0.0416
4	6.02	416	2.0×10^4	0.0208	[0.0189, 0.0229]	0.0213	0.0228
6	7.78	220	4.0×10^4	5.50×10^{-3}	$[4.80 \times 10^{-3}, 6.27 \times 10^{-3}]$	5.77×10^{-3}	7.15×10^{-3}
8	9.03	212	1.4×10^5	1.51×10^{-3}	$[1.32 \times 10^{-3}, 1.73 \times 10^{-3}]$	1.64×10^{-3}	2.34×10^{-3}
10	10.00	205	4.2×10^5	4.88×10^{-4}	$[4.24 \times 10^{-4}, 5.60 \times 10^{-4}]$	4.81×10^{-4}	7.83×10^{-4}
12	10.79	202	1.5×10^6	1.31×10^{-4}	$[1.14 \times 10^{-4}, 1.51 \times 10^{-4}]$	1.43×10^{-4}	2.66×10^{-4}
14	11.46	201	4.9×10^6	4.07×10^{-5}	$[3.53 \times 10^{-5}, 4.67 \times 10^{-5}]$	4.32×10^{-5}	9.14×10^{-5}
16	12.04	200	1.6×10^7	1.25×10^{-5}	$[1.09 \times 10^{-5}, 1.44 \times 10^{-5}]$	1.31×10^{-5}	3.17×10^{-5}
18	12.55	83	2.0×10^7	4.15×10^{-6}	$[3.31 \times 10^{-6}, 5.14 \times 10^{-6}]$	4.02×10^{-6}	1.10×10^{-5}
20	13.01	28	2.0×10^7	1.40×10^{-6}	$[9.30 \times 10^{-7}, 2.02 \times 10^{-6}]$	1.24×10^{-6}	3.87×10^{-6}

the jammer has been converted into noise whose bit-sum standard deviation is only $A\sqrt{M/2}$. The improvement shrinks as M grows because SINR $\approx 2/M$ when $A = M$.

- **Out-of-band jammer.** Without PN the demodulator rejects the jammer through its $\sin(\omega_0/2)$ roll-off. What matters is the peak jammer sum $A|D_M(1)|$ relative to the signal sum M; with $A = M$ this ratio is $|D_M(1)| = 0.28$, 0.55, 2.02 and 0.98 for $M = 50$, 100, 500 and 1000. For $M = 50$ and 100 the jammer can never overturn a bit and the BER is 0. For $M = 500$ it overturns every bit whose jammer phase is near the peak, giving BER $= 0.33$. For $M = 1000$ the ratio is just below 1, so the jammer alone cannot cause an error; the residual BER of 0.013 comes from the noise (standard deviation $\sqrt{MP_n} = 32$) pushing bits that the jammer has brought within $0.02M = 20$ of the threshold across it. With PN the BER is 0.42–0.48 at both jammer frequencies and is therefore *worse* here: spreading takes a jammer the receiver was already rejecting and turns it into in-band noise.

- The simulated and theoretical values agree to within ± 0.007 in every cell, so the behaviour is a property of the system, not of the simulation.

Table 3: BER with the jammer amplitude $A = M$ and $P_n = 1$. Each entry is *simulated (theory)*, with theory from the phase-averaged Equations (7) and (10) (Gaussian approximation for the summed noise).

		In-band, $\omega_0 = 0.002$		Out-of-band, $\omega_0 = 1$	
M	A	No PN	PN	No PN	PN
50	50	0.4932 (0.4934)	0.3744 (0.3681)	0.0000 (0.0000)	0.4208 (0.4208)
100	100	0.4947 (0.4971)	0.4049 (0.4021)	0.0000 (0.0000)	0.4451 (0.4438)
500	500	0.4963 (0.5000)	0.4673 (0.4687)	0.3341 (0.3355)	0.4768 (0.4748)
1000	1000	0.4975 (0.5000)	0.4757 (0.4814)	0.0126 (0.0124)	0.4845 (0.4822)

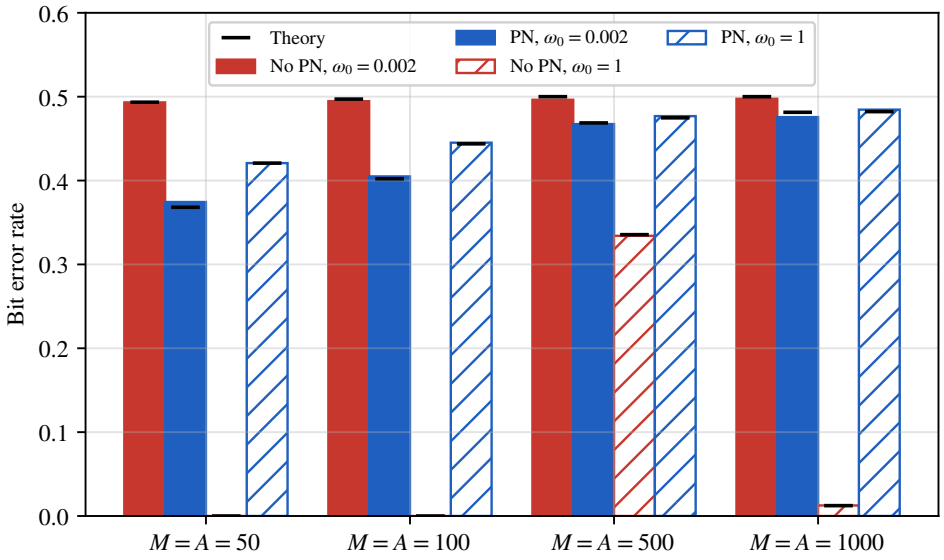


Figure 5: BER at $A = M$; black ticks mark the theoretical values.

5.2 Effect of jammer power

To see the processing gain directly, the in-band jammer amplitude was swept from $A = 0.1$ to 1000 (Figure 6), and the jammer-to-signal ratio at which each system reaches BER $= 10^{-3}$ was read from the theory curves (Table 4).

Table 4: Jammer-to-signal ratio (dB) tolerated at BER $= 10^{-3}$, in-band jammer, $P_n = 1$, and the ratio implied by the brief's condition $A = M$.

M	$10 \log_{10} M$	No PN	PN	Improvement	JSR at $A = M$
50	17.0	-6.5	4.9	11.4	31.0
100	20.0	-5.0	8.3	13.3	37.0
500	27.0	-3.3	16.0	19.2	51.0
1000	30.0	-0.7	19.7	20.5	57.0

Without PN every curve breaks at roughly the same point, JSR ≈ -3 to 0 dB, i.e. $A \approx 1$: once the jammer amplitude exceeds the signal amplitude, errors occur no matter how many samples are integrated. With PN the curves move to the right, and the measured improvement in Table 4 grows from 11.4 dB at $M = 50$ to 20.5 dB at $M = 1000$. This is below the ideal processing gain $10 \log_{10} M$ (17 dB to 30 dB) because a jammer this slow does not average its $\sin^2$ within one bit, so some bits see more interference than others, and because the noise still contributes. The last column shows why the $A = M$ experiment saturates: the jammer is 26 dB to 37 dB stronger than the PN system can tolerate.

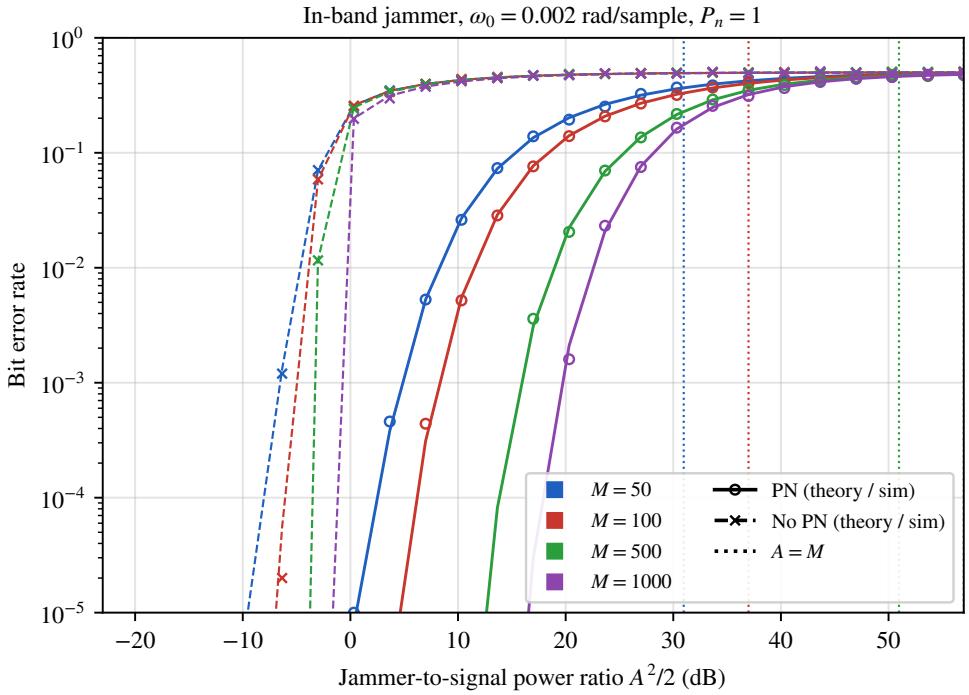


Figure 6: BER against jammer-to-signal power ratio $A^2/2$ for the in-band jammer. Solid lines and circles: PN; dashed lines and crosses: no PN. Dotted verticals mark $A = M$ for each M.

5.3 Effect of jammer frequency

Figure 7 sweeps ω_0 over $0.002\pi \le \omega_0 \le 0.998\pi$ for $M = 100$; the end points are excluded because the brief requires $0 < \omega_0 < \pi$, and at $\omega_0 = \pi$ the sampled jammer $A\sin(\pi n)$ is identically zero. With PN the BER is nearly constant over most of the band and agrees with Equation (10): 0.44 at $A = 100$, and $\mathcal{Q}(\sqrt{100/13.5}) = 3.2 \times 10^{-3}$ at $A = 5$. It departs from these values within roughly $2\pi/M$ of either band edge (falling to about 0.41 at $A = 100$ and rising to about 8×10^{-3} at $A = 5$), where $\sin^2 \omega_0 n$ is no longer averaged within one bit, so the despread interference power varies from bit to bit; the phase-averaged theory reproduces this. Without PN the BER follows the Dirichlet kernel $|D_M(\omega_0)|$: it is 0.5 near DC, falls to zero at every null $\omega_0 = 2\pi\ell/M$, and decays with ω_0. At $A = 5$ the system without PN fails only for $\omega_0 < 0.03\pi$, i.e. inside the main lobe of the demodulator response.

5.4 Spectral view

Figure 8 shows the power spectral densities before and after despreading for $M = 100$, $A = 10$, $\omega_0 = 0.2\pi$. In the channel the spread signal is white with power 1 (0 dB), as is the noise, while the jammer is a line 40 dB above them. After despreading the roles swap: the desired signal collapses back into the narrow sinc^2 spectrum of a rectangular pulse, with 20 dB ($= 10\log_{10} M$) at DC, and the jammer becomes a white floor at $10\log_{10}(A^2/2) = 17$ dB. The integrate-and-dump demodulator is not a sharp filter but a low-pass response, $|D_M(\omega)|$ in (6), whose main lobe covers $|\omega| < 2\pi/M$ and whose sidelobes fall off gradually. That main lobe holds most of the energy of the despread signal (about 90% of a rectangular pulse's sinc^2 spectrum; the rest lies in its sidelobes), while the white jammer contributes only a fraction of order $1/M$ of its power to the bit sum.

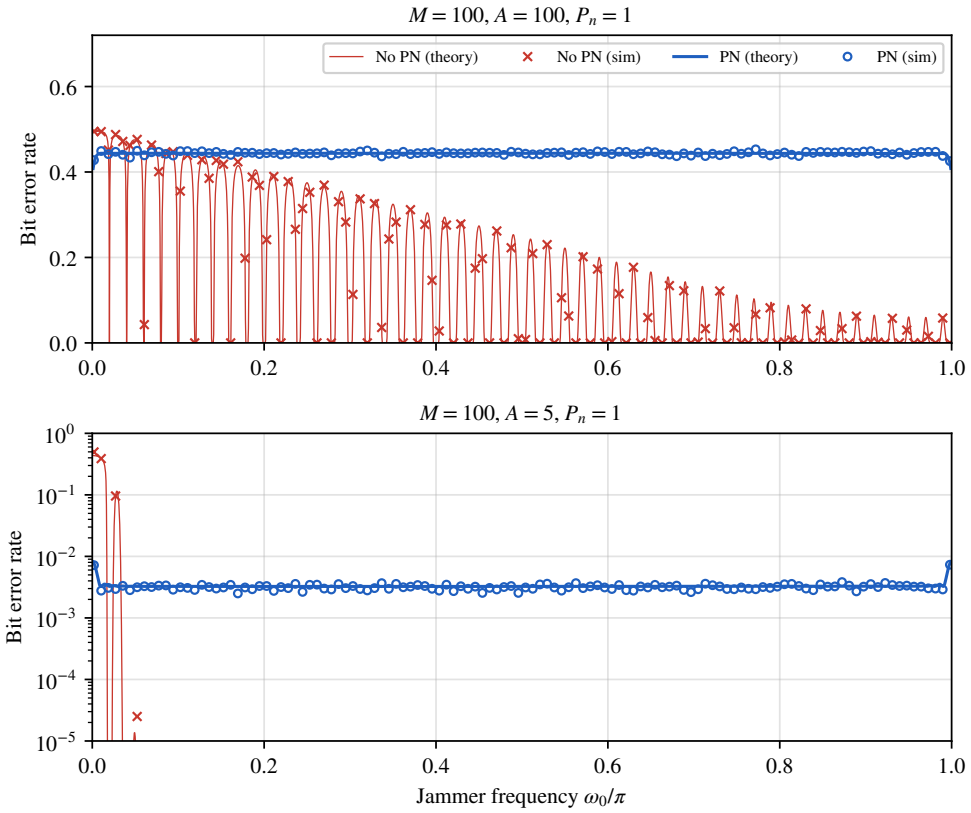


Figure 7: BER against jammer frequency for $M = 100$ and $0 < \omega_0 < \pi$: $A = M$ (top, linear scale) and $A = 5$ (bottom, log scale). Theory curves are the phase-averaged expressions of Section 2.

6 Discussion

6.1 Effect of the PN sequence on the sinusoidal interference

Multiplying by a random ± 1 sequence is a modulation by a wideband signal. In the frequency domain the jammer spectrum, a pair of lines at $\pm\omega_0$, is convolved with the flat spectrum of $p(n)$, so the receiver's despreading multiplier turns the sinusoid into white noise of the same power $A^2/2$ (Figure 8). The same operation applied to the desired signal, which was spread by the same $p(n)$, undoes the spreading because $p^2(n) = 1$. The PN sequence therefore acts as a key that concentrates only the signal back into the narrow band of the demodulator.

6.2 Why the spread-spectrum system outperforms the conventional one

1. **Processing gain.** The demodulator sums M samples. The desired signal adds coherently (amplitude M, power M^2), but the despread jammer adds incoherently (power $MA^2/2$), giving a gain of M in signal-to-jammer ratio. Without PN, an in-band jammer adds coherently as well, so integration gives no gain at all; this is why the conventional system fails at $A \approx 1$ whatever M is. The ideal gain is $10 \log_{10} M = 17\,\text{dB}$ to $30\,\text{dB}$; the measured improvement at BER $= 10^{-3}$ was $11.4\,\text{dB}$ to $20.5\,\text{dB}$ (Table 4).

2. **Weak dependence on the jammer frequency.** Away from the band edges (within about $2\pi/M$ of $\omega_0 = 0$ or π) the PN system's performance depends essentially on the jammer power alone (Figure 7). A jammer cannot find a frequency where the receiver is much weaker; its best strategy is simply more power. In the conventional system the worst case, a jammer placed inside the signal band, is also the easiest for an adversary to choose.

3. **Low probability of interception.** Although not measured here, the spread signal has a

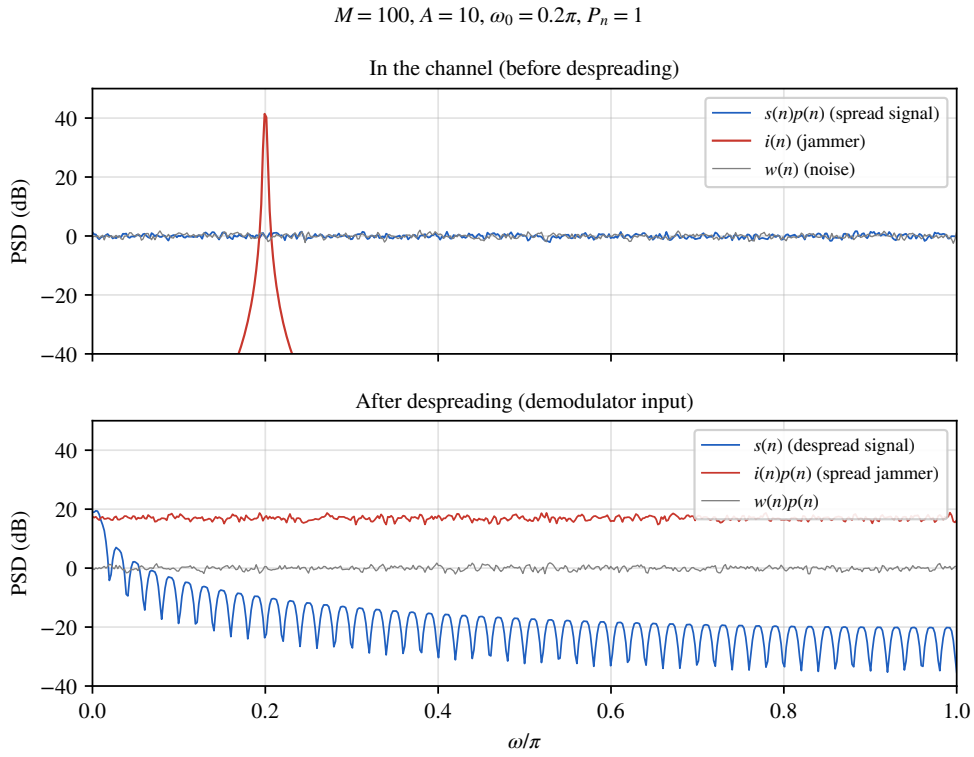


Figure 8: Power spectral densities (Hann-windowed averaged periodogram) in the channel and after despreading.

flat spectrum at the noise level (Figure 8), which makes it difficult to detect or to target, and different PN codes allow several users to share the band (CDMA).

6.3 Interpreting the condition $A \geq M$

The brief asks for $A \geq M$. As derived in Section 2 and confirmed by Table 3, this places the jammer 31 dB to 57 dB above the signal, beyond the ideal $10 \log_{10} M = 17$ dB to 30 dB processing gain, so the PN system operates near BER = 0.5 (0.37–0.48) at both jammer frequencies. The conventional system's result depends strongly on where the jammer sits. In band it is also near 0.5 (0.49–0.50), so PN gives only a modest advantage there. Out of band its integrator rejects the tone, giving BER 0 for $M = 50$ and 100 and 0.013 for $M = 1000$, and only $M = 500$ (0.33), where $|D_M(1)| > 1$, shows substantial errors. In that case the conventional system is clearly better than the PN system. Rather than suggesting that spread spectrum is ineffective, this shows that the benefit is a fixed number of decibels: to obtain a low error rate with $A = M$ the processing gain would need to grow as M^2, which this system's single-PN-chip-per-sample structure cannot provide. Practical systems such as GPS obtain their 43 dB processing gain from a chip rate 2×10^4 times the bit rate.

6.4 Limitations

- Perfect bit and code synchronisation was assumed; in practice code acquisition and tracking are a significant part of a DSSS receiver.
- The PN sequence is generated by a random number generator, as the brief allows. Real systems use deterministic maximal-length or Gold sequences, whose autocorrelation and cross-correlation properties are guaranteed.
- Uniform noise was used as specified; a thermal-noise channel would be Gaussian, for which Equation (4) is exact.

- The Gaussian treatment of the despread jammer in Equation (10) is accurate for large M; at $M = 50$ the simulation and theory still agree within 0.007.

7 Conclusion

Both textbook systems were implemented in MATLAB as one function per block and validated against analytical results. For the binary system with uniform noise the exact Irwin–Hall result lies inside the 95% confidence interval of every simulated BER, from about 0.2 down to 5×10^{-8} (the lowest points rest on few errors and are correspondingly uncertain), and the familiar $\mathcal{Q}(\sqrt{\mathrm{SNR}})$ approximation is pessimistic at high SNR because the noise is bounded. For the spread-spectrum system, the PN sequence converts a narrowband jammer into wideband noise, which the integrate-and-dump demodulator then averages down with an ideal processing gain of M (17 dB to 30 dB). In the simulations this let the receiver tolerate in-band jammers 11.4 dB to 20.5 dB stronger than the conventional receiver at BER $= 10^{-3}$, and made its performance nearly independent of the jammer frequency away from the band edges. Under the brief's condition $A = M$ the jammer power exceeds the ideal processing gain by 14 dB to 27 dB, so the PN system operates near BER $= 0.5$ at both jammer frequencies. The conventional system is also near 0.5 against an in-band jammer, where PN keeps a modest advantage, but against an out-of-band jammer its demodulator filters the tone out (BER 0–0.013 except 0.33 at $M = 500$), and it outperforms the PN system. Jammer frequency therefore matters as much as jammer power for the conventional receiver, but much less for the spread-spectrum one.

Individual contribution

[Describe your own contribution to the group work here: for example which blocks you implemented, which experiments you ran, and which parts of the analysis you led.]

References

[1] V. K. Ingle and J. G. Proakis, *Digital Signal Processing Using MATLAB*, 4th ed. Boston, MA: Cengage Learning, 2017, Projects 12.7 and 12.8.

[2] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York: McGraw-Hill, 2008, ch. 12 (spread-spectrum signals).

[3] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001, ch. 12.

[4] U.S. Space Force, Space Systems Command, *NAVSTAR GPS Space Segment/Navigation User Segment Interfaces*, Interface Specification IS-GPS-200, Rev. N, Aug. 2022. [Online]. Available: `https://www.gps.gov/technical/icwg/`

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A MATLAB code

Block functions

```
function b = bit_generator(N)
% BIT_GENERATOR Binary data generator (Project 12.7, block 1).
% b = bit_generator(N) returns a 1xN row vector of independent,
% equiprobable binary digits (0 or 1).
```

```
    b = double(rand(1, N) >= 0.5);
end
```

Listing 1: bit_generator.m

```
function s = modulator(b, M)
% MODULATOR Rectangular-pulse binary modulator (Project 12.7, block 2).
% s = modulator(b, M) maps every bit 1 to M consecutive +1's and every
% bit 0 to M consecutive -1's, i.e. a sampled antipodal rectangular pulse.
% Output length is M*length(b).
    a = 2*b(:).' - 1; % 0 -> -1, 1 -> +1
    s = reshape(repmat(a, M, 1), 1, []);
end
```

Listing 2: modulator.m

```
function w = noise_generator(L, a)
% NOISE_GENERATOR Uniform additive noise (Project 12.7, block 3).
% w = noise_generator(L, a) returns L i.i.d. samples uniformly
% distributed on (-a, a). The noise power is Pn = a^2/3.
    w = a * (2*rand(1, L) - 1);
end
```

Listing 3: noise_generator.m

```
function y = demodulator(r, M)
% DEMODULATOR Integrate-and-dump demodulator (Project 12.7, block 4).
% y = demodulator(r, M) sums each block of M successive received
% samples. The receiver is assumed to be bit-synchronised, so block k
% is r((k-1)*M+1 : k*M). Returns one decision statistic per bit.
    y = sum(reshape(r, M, []), 1);
end
```

Listing 4: demodulator.m

```
function [b_hat, nerr] = detector(y, b)
% DETECTOR Threshold detector and error counter (Project 12.7, block 5).
% [b_hat, nerr] = detector(y, b) decides 1 when y > 0 and 0 otherwise,
% then counts the decisions that disagree with the transmitted bits b.
% (y == 0 has probability zero for continuous noise.)
    b_hat = double(y > 0);
    nerr = sum(b_hat ~= b);
end
```

Listing 5: detector.m

```
function p = pn_generator(L)
% PN_GENERATOR Binary (+/-1) pseudo-noise sequence (Project 12.8).
% p = pn_generator(L) returns L equiprobable +/-1 chips generated with a
% random number generator. The receiver uses the identical sequence.
    p = 2*double(rand(1, L) >= 0.5) - 1;
end
```

Listing 6: pn_generator.m

```
function i_n = sine_generator(L, A, w0, n0)
% SINE_GENERATOR Sinusoidal interference i(n) = A sin(w0 n) (Project 12.8).
% i_n = sine_generator(L, A, w0) returns L samples starting at n = 0.
% i_n = sine_generator(L, A, w0, n0) starts at n = n0, so a long run can
% be generated in blocks without breaking the phase of the jammer.
    if nargin < 4, n0 = 0; end
    n = n0 + (0:L-1);
    i_n = A * sin(w0 * n);
```

```
end
```

Listing 7: sine_generator.m

System simulator and theory

```
function [ber, nerr, nbits] = binary_comm_sim(M, a, opts)
% BINARY_COMM_SIM Monte-Carlo run of the systems in Figs 12.16 and 12.18.
% [ber, nerr, nbits] = binary_comm_sim(M, a, opts)
% M samples per bit
% a uniform noise half-width (noise on (-a,a), Pn = a^2/3)
% opts struct with optional fields
% .use_pn true to spread with a +/-1 PN sequence (default false)
% .A sinusoidal interference amplitude (default 0)
% .w0 interference frequency in rad/sample, 0 < w0 < pi
% .min_err stop once this many errors are seen (default 200)
% .max_bits hard limit on simulated bits (default 1e6)
% .block bits simulated per block (default 2e4)
% Bits are simulated in blocks so large M does not exhaust memory; the
% jammer phase runs continuously across blocks.
    if nargin < 3, opts = struct(); end
    use_pn = getopt(opts, 'use_pn', false);
    A = getopt(opts, 'A', 0);
    w0 = getopt(opts, 'w0', 0);
    min_err = getopt(opts, 'min_err', 200);
    max_bits = getopt(opts, 'max_bits', 1e6);
    block = getopt(opts, 'block', 2e4);
    if A ~= 0 && ~(w0 > 0 && w0 < pi)
        error('binary_comm_sim:w0', 'Jammer frequency must satisfy 0 < w0 < pi.');
    end
    % keep each block below ~2e7 samples
    block = max(1, min(block, floor(2e7 / M)));

    nerr = 0; nbits = 0; n0 = 0;
    while nerr < min_err && nbits < max_bits
        Nb = min(block, max_bits - nbits);
        L = Nb * M;
        b = bit_generator(Nb); % binary data generator
        s = modulator(b, M); % modulator
        if use_pn
            p = pn_generator(L); % PN sequence generator
            s = s .* p; % spreading
        end
        r = s + noise_generator(L, a); % channel: wideband noise
        if A ~= 0
            r = r + sine_generator(L, A, w0, n0); % channel: sine jammer
        end
        if use_pn
            r = r .* p; % despreading
        end
        y = demodulator(r, M); % demodulator
        [~, e] = detector(y, b); % detector / error counter
        nerr = nerr + e; nbits = nbits + Nb; n0 = n0 + L;
    end
    ber = nerr / nbits;
end

function v = getopt(s, f, d)
    if isfield(s, f), v = s.(f); else, v = d; end
end
```

Listing 8: binary_comm_sim.m

```
function pe = exact_ber_uniform(M, a)
% EXACT_BER_UNIFORM Exact bit error probability for Project 12.7.
% pe = exact_ber_uniform(M, a) returns P(W > M), where W is the sum of M
% i.i.d. Uniform(-a,a) noise samples. By symmetry this is the bit error
% probability of the integrate-and-dump receiver for either bit value.
%
% W = 2a*S - M*a with S ~ Irwin-Hall(M) (sum of M Uniform(0,1)), so
% P(W > M) = P(S > x) = P(S < M - x), x = M(1 + a)/(2a),
% and the Irwin-Hall CDF F(y) = (1/M!) sum_{k=0}^{floor(y)} (-1)^k
% C(M,k) (y-k)^M is evaluated on the short (y <= M/2) side, which keeps
% the alternating sum well conditioned in double precision for M <= 30.
% Works for scalar M and vector a.
    if M > 30
        error('exact_ber_uniform:M', 'Closed form is only well conditioned for M <= 30.');
    end
    pe = zeros(size(a));
    for i = 1:numel(a)
        y = M - M*(1 + a(i)) / (2*a(i)); % = M(a-1)/(2a)
        if y <= 0 % a <= 1: |W| <= M, no errors
            pe(i) = 0; continue;
        end
        k = 0:floor(y);
        logt = gammaln(M+1) - gammaln(k+1) - gammaln(M-k+1) ... % log C(M,k)
             + M*log(y - k) - gammaln(M+1); % (y-k)^M / M!
        logt(y - k <= 0) = -Inf;
        pe(i) = sum((-1).^k .* exp(logt));
    end
end
```

Listing 9: exact_ber_uniform.m

```
function [ber_nopn, ber_pn] = ber_theory_jam(M, A, w0, Pn)
% BER_THEORY_JAM Analytical BER with sinusoidal jamming (Project 12.8).
% [ber_nopn, ber_pn] = ber_theory_jam(M, A, w0, Pn) averages over the
% jammer phase phi seen at the start of a bit (uniform on [0, 2pi)).
% The summed noise is treated as Gaussian with variance M*Pn (CLT).
%
% Without PN the jammer adds a deterministic offset to the bit sum,
% J(phi) = A * sum_{n=0}^{M-1} sin(w0 n + phi),
% so Pe(phi) = [Q((M+J)/s) + Q((M-J)/s)] / 2, s = sqrt(M Pn).
%
% With PN the jammer is multiplied by i.i.d. +/-1 chips, so its sum is
% zero mean with variance A^2 * sum sin^2(w0 n + phi), giving
% Pe(phi) = Q( M / sqrt(M Pn + A^2 sum sin^2(w0 n + phi)) ).
% Scalar M, w0, Pn; A may be a vector.
    phi = 2*pi*(0:359)/360;
    n = (0:M-1).';
    S = sin(w0*n + phi); % M x 360
    J1 = sum(S, 1); % sum sin per phase
    J2 = sum(S.^2, 1); % sum sin^2 per phase
    sig = sqrt(M*Pn);
    ber_nopn = zeros(size(A)); ber_pn = ber_nopn;
    for k = 1:numel(A)
        J = A(k) * J1;
        ber_nopn(k) = mean(0.5*(qfunc_local((M + J)/sig) + qfunc_local((M - J)/sig)));
        ber_pn(k) = mean(qfunc_local(M ./ sqrt(M*Pn + A(k)^2 * J2)));
    end
end
```

Listing 10: ber_theory_jam.m

Experiment scripts

```
%% Project 12.7: Binary data communications system (Ingle & Proakis 4e)
% Simulates the system of Fig. 12.16:
% bit generator -> modulator (M samples/bit) -> + uniform noise ->
% integrate-and-dump demodulator -> detector & error counter
% and measures the bit error rate (BER) against SNR = M / Pn, Pn = a^2/3.
%
% Two sweeps are run, as suggested in the brief:
% (a) Pn fixed (= 1) while M is changed,
% (b) M fixed (= 10) while Pn (i.e. a) is changed.
% Results are compared with the Gaussian (CLT) approximation
% Pe ~ Q(sqrt(M/Pn)) and with the exact result from exact_ber_uniform.
clear; close all; clc;
rng(12); % reproducible results
if exist('OCTAVE_VERSION', 'builtin'), graphics_toolkit('gnuplot'); end

min_err = 200; % stop after this many errors
max_bits = 2e7; % ... or this many bits

%% Illustration of the signals through each block
M = 10; a = sqrt(3); % Pn = 1, SNR = 10 dB
b = bit_generator(8);
s = modulator(b, M);
w = noise_generator(numel(s), a);
r = s + w;
y = demodulator(r, M);
b_hat = detector(y, b);
n = 0:numel(s)-1;

f = figure('visible', 'off');
subplot(3,1,1); stairs(n, s, 'LineWidth', 1.2); ylim([-1.5 1.5]); grid on;
title(sprintf('Modulator output s(n), bits = [%s]', num2str(b)));
ylabel('s(n)');
subplot(3,1,2); plot(n, r, '.-'); grid on;
title(sprintf('Received r(n) = s(n) + w(n), a = %.2f (P_n = %.2f)', a, a^2/3));
ylabel('r(n)');
subplot(3,1,3); stem(0:numel(y)-1, y, 'filled'); hold on;
plot([-0.5 numel(y)-0.5], [0 0], 'k--'); grid on;
title(sprintf('Demodulator output y(k), detected bits = [%s]', num2str(b_hat)));
xlabel('bit index k'); ylabel('y(k)');
save_fig(f, 'p127_waveforms');

%% (a) Pn fixed, M varied
a_fix = sqrt(3); Pn_fix = a_fix^2/3; % Pn = 1
M_list = [1 2 3 4 6 8 10 12 14 16 18 20];
ber_a = zeros(size(M_list)); nb_a = ber_a; ne_a = ber_a;
for k = 1:numel(M_list)
    [ber_a(k), ne_a(k), nb_a(k)] = binary_comm_sim(M_list(k), a_fix, ...
        struct('min_err', min_err, 'max_bits', max_bits));
    fprintf('(a) M = %2d SNR = %5.2f dB BER = %.3e (%d bits)\n', ...
        M_list(k), 10*log10(M_list(k)/Pn_fix), ber_a(k), nb_a(k));
end
snr_a = M_list / Pn_fix;
gauss_a = qfunc_local(sqrt(snr_a));
exact_a = zeros(size(M_list));
for k = 1:numel(M_list), exact_a(k) = exact_ber_uniform(M_list(k), a_fix); end

%% (b) M fixed, Pn varied
M_fix = 10;
snr_db = 0:1:13;
Pn_b = M_fix ./ 10.^(snr_db/10);
a_b = sqrt(3*Pn_b);
ber_b = zeros(size(snr_db)); nb_b = ber_b; ne_b = ber_b;
for k = 1:numel(snr_db)
    [ber_b(k), ne_b(k), nb_b(k)] = binary_comm_sim(M_fix, a_b(k), ...
```

```
        struct('min_err', min_err, 'max_bits', max_bits));
    fprintf('(b) a = %5.2f SNR = %5.2f dB BER = %.3e (%d bits)\n', ...
        a_b(k), snr_db(k), ber_b(k), nb_b(k));
end
gauss_b = qfunc_local(sqrt(10.^(snr_db/10)));
exact_b = exact_ber_uniform(M_fix, a_b);

%% Plot BER vs SNR for both sweeps
snr_fine = linspace(-1, 14, 200);
f = figure('visible', 'off');
semilogy(10*log10(snr_a), ber_a, 'o', 'MarkerSize', 7, 'LineWidth', 1.2); hold on;
semilogy(snr_db, ber_b, 's', 'MarkerSize', 7, 'LineWidth', 1.2);
semilogy(snr_fine, qfunc_local(sqrt(10.^(snr_fine/10))), 'k-', 'LineWidth', 1.2);
semilogy(10*log10(snr_a), exact_a, 'b--', 'LineWidth', 1);
semilogy(snr_db, exact_b, 'r:', 'LineWidth', 1.4);
grid on; xlim([-1 14]); ylim([1e-8 1]);
xlabel('SNR = M / P_n (dB)'); ylabel('Bit error rate');
title('Project 12.7: BER of the binary system with uniform noise');
legend('Sim. (a): P_n = 1, M varied', 'Sim. (b): M = 10, P_n varied', ...
       'Gaussian approx. Q(\surd SNR)', 'Exact (a)', 'Exact (b)', ...
       'Location', 'southwest');
save_fig(f, 'p127_ber');

%% Save numeric results
res127 = struct('M_list', M_list, 'snr_a_db', 10*log10(snr_a), 'ber_a', ber_a, ...
    'nbits_a', nb_a, 'nerr_a', ne_a, 'exact_a', exact_a, 'gauss_a', gauss_a, ...
    'M_fix', M_fix, 'snr_b_db', snr_db, 'a_b', a_b, 'ber_b', ber_b, ...
    'nbits_b', nb_b, 'nerr_b', ne_b, 'exact_b', exact_b, 'gauss_b', gauss_b, ...
    'wave_b', b, 'wave_s', s, 'wave_r', r, 'wave_y', y, 'wave_bhat', b_hat, 'wave_M', 10);
save(fullfile(fileparts(mfilename('fullpath')), '..', 'results', 'res127.mat'), 'res127', '-v7');
write_table_127(res127);
```

Listing 11: project12_7.m

```
%% Project 12.8: Binary spread-spectrum communications (Ingle & Proakis 4e)
% Simulates the system of Fig. 12.18: the Project 12.7 system with the
% modulator output multiplied by a +/-1 PN sequence p(n), a channel that
% adds wideband noise w(n) and a sinusoidal jammer i(n) = A sin(w0 n),
% and a receiver that multiplies by the same p(n) before demodulating.
%
% Experiments
% 1. Brief: BER with and without PN for M = 50, 100, 500, 1000, A = M,
% for an in-band (low-frequency) and an out-of-band jammer.
% 2. BER against jammer amplitude A (jammer-to-signal ratio) per M.
% 3. BER against jammer frequency w0 for fixed M.
% 4. Power spectra before and after despreading.
clear; close all; clc;
rng(128);
if exist('OCTAVE_VERSION', 'builtin'), graphics_toolkit('gnuplot'); end

a = sqrt(3); Pn = a^2/3; % wideband noise, Pn = 1
M_list = [50 100 500 1000];
w_in = 0.002; % rad/sample, inside 2*pi/M for all M
w_out = 1; % rad/sample (~0.318*pi)
min_err = Inf; % experiment 1: fixed bit budget
bits_for = @(M) max(2e4, round(2e7/M)); % ~2e7 samples per point

%% 1. Brief experiment: A = M
fprintf('Experiment 1: A = M, Pn = %.2f\n', Pn);
w_set = [w_in w_out];
T = zeros(numel(M_list), 2*numel(w_set)); % simulated BERs
Tth = T; % theory
for i = 1:numel(M_list)
    M = M_list(i); A = M;
    for j = 1:numel(w_set)
```

```
        o = struct('A', A, 'w0', w_set(j), 'min_err', min_err, 'max_bits', bits_for(M));
        o.use_pn = false; ber0 = binary_comm_sim(M, a, o);
        o.use_pn = true; ber1 = binary_comm_sim(M, a, o);
        [t0, t1] = ber_theory_jam(M, A, w_set(j), Pn);
        T(i, 2*j-1:2*j) = [ber0 ber1];
        Tth(i, 2*j-1:2*j) = [t0 t1];
        fprintf(' M = %4d w0 = %.3f no PN: %.4f (th %.4f) PN: %.4f (th %.4f)\n', ...
            M, w_set(j), ber0, t0, ber1, t1);
    end
end

f = figure('visible', 'off');
semilogx(M_list, T(:,1), 'ro-', M_list, T(:,2), 'bo-', ...
         M_list, T(:,3), 'rs--', M_list, T(:,4), 'bs--', 'LineWidth', 1.2, 'MarkerSize', 7);
grid on; ylim([0 0.6]); xlim([40 1200]);
xlabel('Samples per bit M (with A = M)'); ylabel('Bit error rate');
title('Project 12.8: BER with jammer amplitude A = M, P_n = 1');
legend(sprintf('No PN, \\omega_0 = %.3f', w_in), sprintf('PN, \\omega_0 = %.3f', w_in), ...
       sprintf('No PN, \\omega_0 = %g', w_out), sprintf('PN, \\omega_0 = %g', w_out), ...
       'Location', 'southwest');
save_fig(f, 'p128_ber_A_eq_M');

%% 2. BER against jammer amplitude (in-band jammer)
A_list = logspace(-1, 3, 25);
jsr_db = 10*log10(A_list.^2/2); % jammer power / signal power per sample
B0 = nan(numel(M_list), numel(A_list)); B1 = B0; Th0 = B0; Th1 = B0;
for i = 1:numel(M_list)
    M = M_list(i);
    [Th0(i,:), Th1(i,:)] = ber_theory_jam(M, A_list, w_in, Pn);
    for k = 1:numel(A_list)
        o = struct('A', A_list(k), 'w0', w_in, 'min_err', 1000, ...
                   'max_bits', round(bits_for(M)/4));
        o.use_pn = false; B0(i,k) = binary_comm_sim(M, a, o);
        o.use_pn = true; B1(i,k) = binary_comm_sim(M, a, o);
    end
    fprintf('Experiment 2: M = %d done\n', M);
end

f = figure('visible', 'off');
cols = {'b', 'r', [0 0.6 0], 'm'};
if exist('gobjects') > 0 % MATLAB: graphics-handle array
    h = gobjects(1, numel(M_list));
else % Octave: handles are doubles
    h = zeros(1, numel(M_list));
end
for i = 1:numel(M_list)
    h(i) = semilogy(jsr_db, max(B1(i,:), 1e-6), 'o', 'Color', cols{i}, 'MarkerSize', 5); hold on;
    semilogy(jsr_db, max(Th1(i,:), 1e-6), '-', 'Color', cols{i}, 'LineWidth', 1.1);
    semilogy(jsr_db, max(B0(i,:), 1e-6), 'x', 'Color', cols{i}, 'MarkerSize', 6);
    semilogy(jsr_db, max(Th0(i,:), 1e-6), '--', 'Color', cols{i}, 'LineWidth', 0.8);
    jA = 10*log10(M_list(i)^2/2); % A = M for this M
    semilogy([jA jA], [1e-6 1], '-.', 'Color', cols{i}, 'LineWidth', 0.6);
end
grid on; ylim([1e-5 1]); xlim([min(jsr_db) max(jsr_db)]);
xlabel('Jammer-to-signal power ratio A^2/2 (dB)'); ylabel('Bit error rate');
title(sprintf('BER vs jammer power, \\omega_0 = %.3f rad/sample (o PN, x no PN)', w_in));
legend(h, arrayfun(@(m) sprintf('M = %d', m), M_list, 'UniformOutput', false), ...
       'Location', 'southeast');
save_fig(f, 'p128_ber_vs_jsr');

%% 3. BER against jammer frequency
M = 100;
% The brief requires 0 < w0 < pi. At w0 = pi the sampled jammer
% A*sin(pi*n) is identically zero, so both end points are excluded.
```

```
w_list = pi * linspace(0.002, 0.998, 120); % simulated points
w_fine = pi * linspace(0.001, 0.999, 2000);% theory curve
A_pair = [M 5];
W0 = zeros(2, numel(w_list)); W1 = W0;
Wt0 = zeros(2, numel(w_fine)); Wt1 = Wt0;
for q = 1:2
    for k = 1:numel(w_list)
        o = struct('A', A_pair(q), 'w0', w_list(k), 'min_err', 500, 'max_bits', 4e4);
        o.use_pn = false; W0(q,k) = binary_comm_sim(M, a, o);
        o.use_pn = true; W1(q,k) = binary_comm_sim(M, a, o);
    end
    for k = 1:numel(w_fine)
        [Wt0(q,k), Wt1(q,k)] = ber_theory_jam(M, A_pair(q), w_fine(k), Pn);
    end
end
fprintf('Experiment 3 done\n');

f = figure('visible', 'off');
floor_ = 1e-5;
for q = 1:2
    subplot(2,1,q);
    h1 = semilogy(w_fine/pi, max(Wt0(q,:), floor_), 'r-', 'LineWidth', 0.8); hold on;
    h2 = semilogy(w_list/pi, max(W0(q,:), floor_), 'rx', 'MarkerSize', 5);
    h3 = semilogy(w_fine/pi, max(Wt1(q,:), floor_), 'b-', 'LineWidth', 1.4);
    h4 = semilogy(w_list/pi, max(W1(q,:), floor_), 'bo', 'MarkerSize', 4);
    grid on; ylim([floor_ 1]);
    title(sprintf('M = %d, A = %g, P_n = 1', M, A_pair(q)));
    ylabel('BER');
    if q == 1
        legend([h1 h2 h3 h4], 'No PN (theory)', 'No PN (sim)', 'PN (theory)', 'PN (sim)', ...
               'Location', 'southwest');
    end
end
xlabel('\omega_0 / \pi');
save_fig(f, 'p128_ber_vs_w0');

%% 4. Spectra before and after despreading
M = 100; A = 10; w0 = 0.2*pi; Nb = 400; L = Nb*M; Nfft = 1024;
b = bit_generator(Nb);
s = modulator(b, M);
p = pn_generator(L);
jam = sine_generator(L, A, w0);
w = noise_generator(L, a);
r = s.*p + jam + w; % received (spread) signal
z = r .* p; % after despreading
[Psp, om] = psd_avg(s.*p, Nfft); % transmitted spread signal
Pj = psd_avg(jam, Nfft); % jammer in the channel
Pw = psd_avg(w, Nfft); % noise in the channel
Ps = psd_avg(s, Nfft); % desired signal after despreading
Pjd = psd_avg(jam.*p, Nfft); % jammer after despreading
Pwd = psd_avg(w.*p, Nfft); % noise after despreading
Pz = psd_avg(z, Nfft); % total after despreading
dB = @(P) 10*log10(P + 1e-6);

f = figure('visible', 'off');
subplot(2,1,1);
h1 = plot(om/pi, dB(Psp), 'b'); hold on;
h2 = plot(om/pi, dB(Pj), 'r');
h3 = plot(om/pi, dB(Pw), 'Color', [0.5 0.5 0.5]);
grid on; ylim([-40 60]); ylabel('PSD (dB)');
title('In the channel (before despreading)');
legend([h1 h2 h3], 's(n)p(n)', 'i(n)', 'w(n)', 'Location', 'northeast');
subplot(2,1,2);
h1 = plot(om/pi, dB(Ps), 'b'); hold on;
```

```
h2 = plot(om/pi, dB(Pjd), 'r');
h3 = plot(om/pi, dB(Pwd), 'Color', [0.5 0.5 0.5]);
grid on; ylim([-40 60]); ylabel('PSD (dB)'); xlabel('\omega / \pi');
title('After despreading (input to the demodulator)');
legend([h1 h2 h3], 's(n)', 'i(n)p(n)', 'w(n)p(n)', 'Location', 'northeast');
save_fig(f, 'p128_spectra');

%% Save numeric results
res128 = struct('M_list', M_list, 'w_in', w_in, 'w_out', w_out, 'Pn', Pn, ...
    'T', T, 'Tth', Tth, 'A_list', A_list, 'jsr_db', jsr_db, 'B0', B0, 'B1', B1, ...
    'Th0', Th0, 'Th1', Th1, 'w_list', w_list, 'A_pair', A_pair, ...
    'W0', W0, 'W1', W1, 'Wt0', Wt0, 'w_fine', w_fine, 'Wt1', Wt1, ...
    'psd_om', om, 'Psp', Psp, 'Pj', Pj, 'Pw', Pw, 'Ps', Ps, 'Pjd', Pjd, 'Pwd', Pwd, ...
    'psd_M', M, 'psd_A', A, 'psd_w0', w0);
save(fullfile(fileparts(mfilename('fullpath')), '..', 'results', 'res128.mat'), 'res128', '-v7');
write_table_128(res128);
```

Listing 12: project12_8.m